CODING SHUTTLE

Kubernetes

Homework

Recap

1. Introduction to Kubernetes
2. Installing Kubernetes Cluster using Minikube
3. Pods, Namespace, Deployment, ReplicaSet
4. Services
5. Persistent Volumes
6. Stateful Set

Homework

1. Deploy a Simple Web Application: Using Minikube, create a deployment for a basic web application like Nginx. Expose the application as a service to access it from your host machine. Scale the deployment to 3 replicas and ensure all replicas are reachable.
2. Namespace Isolation: Create two namespaces, dev and prod. Deploy a pod in each namespace with an environment variable distinguishing them (e.g., "ENV=dev" in dev and "ENV=prod" in prod). Verify that the pods are isolated by namespace.
3. Persistent Volumes with StatefulSet: Set up a StatefulSet with a sample database, such as MySQL. Attach a PersistentVolume to each pod to ensure data persists across restarts. Test by deleting and recreating pods to confirm data persistence.

Homework

4. StatefulSet with Headless Service
    - Deploy a StatefulSet for a simple application (e.g., a Redis instance or MySQL instance) that requires stable, unique network IDs.
    - Create a headless service to enable direct pod-to-pod communication.
    - Verify that each pod has a stable hostname and consistent storage.